THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	$a_2 - a_1 = 4$
2	**(a)** 5	$\sqrt{3^2 + 4^2} = 5$
3	**(a)** 5	$\sqrt{x^2 + y^2}$
4	**(a)** 39	$a + (n-1)d = 39$
5	**(b)** 3	Perimeter ratio = side ratio
6	**(c)** $(4,\ 6)$	Average the coordinates
7	**(a)** 55	$S_n = \frac{n}{2}[2a + (n-1)d]$
8	**(a)** 6	$\frac{AD}{DB} = \frac{AE}{EC}$
9	**(c)** first quadrant	Signs of coordinates
10	**(b)** $4, 7, 10, 13, \ldots$	Only $4, 7, 10, 13, \ldots$ has the same difference 3 between consecutive terms
11	**(b)** 3	Side ratio is the square root of the area ratio
12	**(a)** $(3,\ 0)$	$y = 0$ on the x-axis
13	**(c)** 8	$a_2 = a + d$
14	**(d)** 5	Pythagoras
15	**(c)** 10	Distance formula

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$5, 8, 11, 14, \ldots$ (2).	2
17	$h^2 = 36 + 64 = 100$ (1); $h = 10$ cm (1).	1+1
18	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(3)^2 + (4)^2} = \sqrt{25}$ (1); $= 5$ units (1).	1+1+1
19	$d = a_2 - a_1 = 11 - (6)$ (1) $= 5$ (1).	1+1
20	Area ratio $= (2/3)^2$ (1) $= 4 : 9$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$a = 3$, $d = 4$ (1); $a_n = a + (n-1)d$ (1); $a_{12} = 3 + 11(4) = 47$ (1).	1+1+1
22	$h^2 = 3^2 + 4^2 = 25$ (1); $h = 5$ cm (1); perimeter $= 12$ cm (1).	1+1+1
23	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(3)^2 + (4)^2} = \sqrt{25}$ (1); $= 5$ units (1).	1+1+1
24	$a = 2$, $d = 3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{10} = \frac{10}{2}[4 + 9(3)] = 155$ (1).	1+1+1
	OR: By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{2}{3} = \frac{4}{EC}$ (1); $EC = 6$ cm (1).	1+1+1
25	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (5,\ 5)$ (1).	1+1
	OR: $a_n = a + (n-1)d$: $54 = 4 + (n-1)(5)$ (1); $n - 1 = 10$ (1); $n = 11$, so it is the 11th term (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$a = 7$, $d = 3$, $l = 64$ (1); $l = a + (n-1)d$ gives $n - 1 = 19$ (1); $n = 20$ (1).	1+1+1
	OR: Ladder is the hypotenuse of a right triangle (1); $d^2 = 5^2 - 4^2 = 9$ (1); $d = 3$ m (1).	1+1+1
27	Area $= \frac{1}{2}\lvert x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\rvert$ (1); $= \frac{1}{2}\lvert 0 + 12 + 0\rvert$ (1); $= 6$ sq. units (1).	1+1+1
	OR: $a_n = 6 + 19(4) = 82$ (1); $S_n = \frac{20}{2}(6 + 82) = 880$ (2).	1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 3	$a_3 = a + 2d$
2	**(b)** equal	AAA similarity
3	**(b)** 5	Abscissa is the x-coordinate
4	**(a)** -3	$a_2 - a_1 = -3$
5	**(b)** 13	$\sqrt{5^2 + 12^2} = 13$
6	**(d)** 13	$\sqrt{x^2 + y^2}$
7	**(d)** 26	$a + (n-1)d = 26$
8	**(d)** 5	Perimeter ratio = side ratio
9	**(d)** $(4,\ 3)$	Average the coordinates
10	**(c)** 72	$S_n = \frac{n}{2}[2a + (n-1)d]$
11	**(b)** 8	$\frac{AD}{DB} = \frac{AE}{EC}$
12	**(c)** second quadrant	Signs of coordinates
13	**(d)** $9, 7, 5, 3, \ldots$	Only $9, 7, 5, 3, \ldots$ has the same difference -2 between consecutive terms
14	**(d)** 5	Side ratio is the square root of the area ratio
15	**(d)** $(-4,\ 0)$	$y = 0$ on the x-axis

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (5,\ 8)$ (1).	1+1
17	$d = 9$ constant (1); next term 38 (1).	1+1
18	$\frac{AD}{DB} = \frac{AE}{EC}$ (1); $EC = 8$ cm (1).	1+1
19	Quadrant I (1); reflection $(3, -5)$ (1).	1+1
20	$10, 8, 6, 4, \ldots$ (2).	2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Largest side 10: $(10)^2 = 100$ (1); sum of squares of the others $= 100$ (1); equal, so by the converse of Pythagoras it is right-angled at the vertex opposite the longest side (1).	1+1+1
22	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{1(6)+2(0)}{3} = 2$ (1); $y = 3$ (1).	1+1+1
23	$a = 5$, $d = 3$ (1); $a_n = a + (n-1)d$ (1); $a_{10} = 5 + 9(3) = 32$ (1).	1+1+1
24	$h^2 = 5^2 + 12^2 = 169$ (1); $h = 13$ cm (1); perimeter $= 30$ cm (1).	1+1+1
	OR: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(8)^2 + (6)^2} = \sqrt{100}$ (1); $= 10$ units (1).	1+1+1
25	$a = 5$, $d = 5$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_8 = \frac{8}{2}[10 + 7(5)] = 180$ (1).	1+1+1
	OR: By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{3}{5} = \frac{6}{EC}$ (1); $EC = 10$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	The triangles formed are similar (AA) (1); $\frac{H}{6} = \frac{30}{4}$ (1); $H = 45$ m (1).	1+1+1
	OR: Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (2,\ 3)$ (2).	1+2
27	$a = 5$, $d = 5$, $l = 80$ (1); $l = a + (n-1)d$ gives $n - 1 = 15$ (1); $n = 16$ (1).	1+1+1
	OR: Ladder is the hypotenuse of a right triangle (1); $d^2 = 13^2 - 12^2 = 25$ (1); $d = 5$ m (1).	1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	$a_2 = a + d$
2	**(b)** 13	Pythagoras
3	**(c)** 13	Distance formula
4	**(a)** 5	$a_3 = a + 2d$
5	**(b)** $3^2 + 4^2 = 5^2$	Converse of Pythagoras
6	**(a)** 7	Abscissa is the x-coordinate
7	**(c)** 5	$a_2 - a_1 = 5$
8	**(b)** 17	$\sqrt{8^2 + 15^2} = 17$
9	**(d)** 17	$\sqrt{x^2 + y^2}$
10	**(a)** 37	$a + (n-1)d = 37$
11	**(b)** 4	Perimeter ratio = side ratio
12	**(d)** $(1,\ 5)$	Average the coordinates
13	**(d)** 105	$S_n = \frac{n}{2}[2a + (n-1)d]$
14	**(c)** 15	$\frac{AD}{DB} = \frac{AE}{EC}$
15	**(c)** third quadrant	Signs of coordinates

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$h^2 = 81 + 144 = 225$ (1); $h = 15$ cm (1).	1+1
17	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(3)^2 + (4)^2} = \sqrt{25}$ (1); $= 5$ units (1).	1+1+1
18	$d = a_2 - a_1 = 17 - (20)$ (1) $= -3$ (1).	1+1
19	Area ratio $= (3/4)^2$ (1) $= 9 : 16$ (1).	1+1
20	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (1,\ 1)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (-1,\ 3)$ (1).	1+1
22	$a_n = a + (n-1)d$: $59 = 3 + (n-1)(7)$ (1); $n - 1 = 8$ (1); $n = 9$, so it is the 9th term (1).	1+1+1
23	Largest side 9: $(9)^2 = 81$ (1); sum of squares of the others $= 74$ (1); not equal, so it is not right-angled (1).	1+1+1
24	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{1(8)+1(2)}{2} = 5$ (1); $y = 4$ (1).	1+1+1
	OR: $a = 2$, $d = 7$ (1); $a_n = a + (n-1)d$ (1); $a_8 = 2 + 7(7) = 51$ (1).	1+1+1
25	$h^2 = 8^2 + 15^2 = 289$ (1); $h = 17$ cm (1); perimeter $= 40$ cm (1).	1+1+1
	OR: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(5)^2 + (12)^2} = \sqrt{169}$ (1); $= 13$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Area $= \frac{1}{2}\lvert x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\rvert$ (1); $= \frac{1}{2}\lvert -5 + 25 + 0\rvert$ (1); $= 10$ sq. units (1).	1+1+1
	OR: $a_n = 9 + 13(3) = 48$ (1); $S_n = \frac{14}{2}(9 + 48) = 399$ (2).	1+2
27	The triangles formed are similar (AA) (1); $\frac{H}{5} = \frac{21}{3}$ (1); $H = 35$ m (1).	1+1+1
	OR: Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (4,\ 4)$ (2).	1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.